



International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering (CCE)



LAB Report #06



Course Title: OOP with JAVA Programming Sessional
Course Code: CCE-1208

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Task name: Demonstration of Inheritance (Multilevel & Hierarchical).

Task 1: Grandfather→Father → Son (Multilevel Inheritance).

- Create a class Grandfather with a method company().
- Create a class Father that extends Grandfather and adds a method car().
- Create a class Son that extends Father and add a method.
- Demonstrate how Son can access all properties.

Objective(s):

- To understand multilevel inheritance in Java.
- To learn how a subclass can access methods of its parent and grandparent classes.
- To practice inheritance and method calling in object-oriented programming.

Problem Analysis: The program should create three classes:

- Grandfather with a method company()
- Father that extends Grandfather and adds a method car()
- Son that extends Father and adds another method

The Son class should be able to access all methods from Grandfather, Father, and itself.

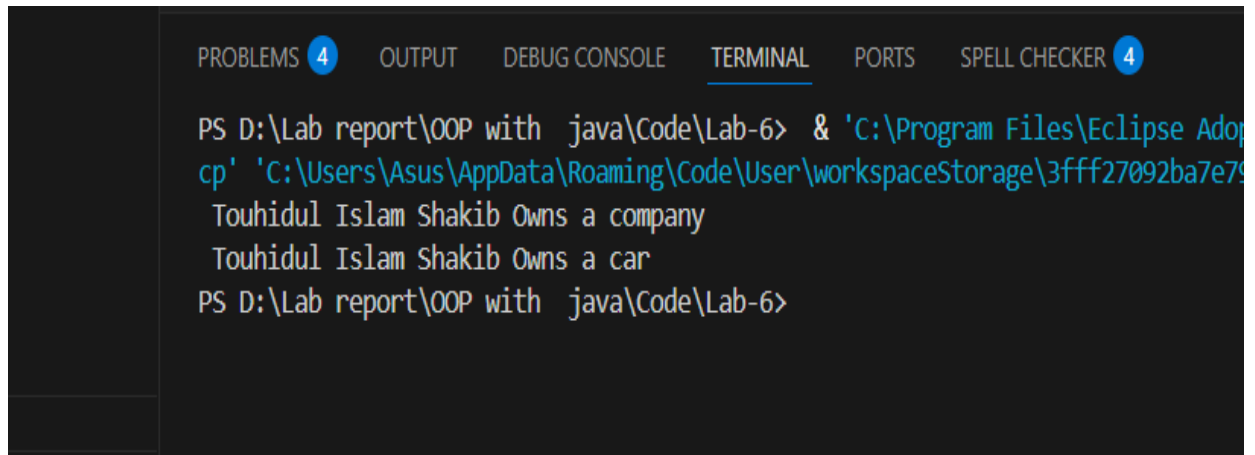
Process:

- Create a Grandfather class with a company() method.
- Create a Father class that extends Grandfather and includes a car() method.
- Create a Son class that extends Father and adds its own method.
- Create an object of Son.
- Call all methods using the Son object.

Source Code:

<pre>class GrandFather { String name = "Ruhul Amin"; void company() { System.out.println(name + " Owns a company"); } } class Father extends GrandFather { public Father() { this.name = "MD.Shahjahan"; } void car() { System.out.println(name + " Owns a car"); } }</pre>	<pre>class Son extends Father { public Son(String n) { this.name = n; } public class FamilyDetails { public static void main(String[] args) { Son s = new Son("Touhidul Islam Shakib"); s.company(); s.car(); } } }</pre>
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Output:

A screenshot of the Eclipse IDE's integrated terminal. The terminal has a dark background with light-colored text. At the top, there is a tab bar with several tabs: 'PROBLEMS' (with a blue circle containing the number 4), 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is the active tab and underlined), 'PORTS', and 'SPELL CHECKER' (with a blue circle containing the number 4). The terminal content shows a command prompt 'PS D:\Lab report\OOP with java\Code\Lab-6>' followed by a command to run a Java program. The output of the program is displayed on the next two lines: 'Touhidul Islam Shakib Owns a company' and 'Touhidul Islam Shakib Owns a car'. The prompt 'PS D:\Lab report\OOP with java\Code\Lab-6>' appears again at the bottom of the terminal.

```
PS D:\Lab report\OOP with java\Code\Lab-6> & 'C:\Program Files\Eclipse AdoptOpenJDK\jdk-11.0.10-hotspot\bin\java.exe' -cp 'C:\Users\Asus\AppData\Roaming\Code\User\workspaceStorage\3fff27092ba7e79...
Touhidul Islam Shakib Owns a company
Touhidul Islam Shakib Owns a car
PS D:\Lab report\OOP with java\Code\Lab-6>
```

Discussion & Conclusion:

This program demonstrates the concept of Multilevel Inheritance in Java. In this experiment, the Father class inherits the properties and methods of the Grandfather class, and the Son class inherits from the Father class. As a result, the Son class can access the methods of both Grandfather and Father classes. Through this experiment, we learned how inheritance works through multiple levels and how properties and methods can be reused in different classes. We also gained practical knowledge about superclass and subclass relationships, method inheritance, and code reusability in Object-Oriented Programming (OOP). The program was executed successfully and produced the expected output, which helped us understand the practical implementation and importance of multilevel inheritance in Java.

Task 2: Person→Doctor/Teacher/Engineer (Hierarchical Inheritance)

- Create a superclass Person with method displayInfo().
- Create subclasses Doctor, Teacher, and Engineer extending Person.
- Each subclass will have its own method to describe their work.

Objective(s):

- To understand hierarchical inheritance in Java.
- To learn how one superclass can be inherited by multiple subclasses.
- To practice using methods in different child classes.

Problem Analysis: The program should create:

- A superclass Person with method displayInfo()
- Three subclasses: Doctor, Teacher, Engineer

Each subclass should extend Person and include its own method to describe its work.

Process:

- Create a Person class with a displayInfo() method.
- Create three subclasses: Doctor, Teacher, and Engineer.
- Each subclass should have its own working method.
- Create objects of all three subclasses.
- Call both the inherited method and the subclass method.

Source Code:

```
class Person {
void displayInfo() {
System.out.println("This is a person.");
}
}
class Doctor extends Person {
void work() {
System.out.println("Doctor treats patients.");
}
}
class Teacher extends Person {
void work() {
System.out.println("Teacher teaches students.");
}
}

class Engineer extends Person {
void work() {
System.out.println("Engineer develops software
and machines.");
}
}
```

```
public class Main {
public static void main(String[] args) {

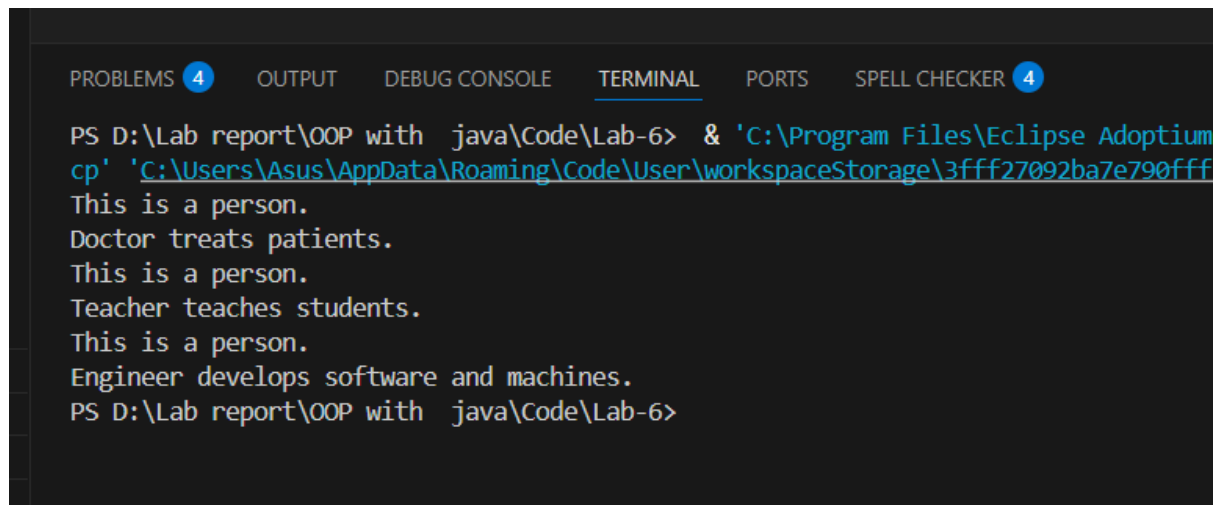
Doctor d = new Doctor();
Teacher t = new Teacher();
Engineer e = new Engineer();

d.displayInfo();
d.work();

t.displayInfo();
t.work();

e.displayInfo();
e.work();
}
}
```

Output:



```
PS D:\Lab report\OOP with java\Code\Lab-6> & 'C:\Program Files\Eclipse Adoptium
cp' 'C:\Users\Asus\AppData\Roaming\Code\User\workspaceStorage\3fff27092ba7e790fff
This is a person.
Doctor treats patients.
This is a person.
Teacher teaches students.
This is a person.
Engineer develops software and machines.
PS D:\Lab report\OOP with java\Code\Lab-6>
```

Discussion & Conclusion:

This program demonstrates the concept of Hierarchical Inheritance in Java. In this experiment, a superclass named Person is created, and multiple subclasses such as Doctor, Teacher, and Engineer inherit the common properties and methods of the superclass. Each subclass also contains its own specific method to describe its work. Through this experiment, we learned how multiple subclasses can share common features from a single superclass while maintaining their own individual behaviors. We also gained practical knowledge about inheritance, method usage, and code organization in Object-Oriented Programming (OOP). The program was executed successfully and produced the expected output, which helped us understand the practical implementation and importance of hierarchical inheritance in Java.

